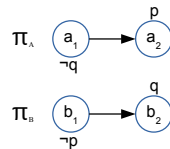


Table 3 represents the game in normal-form of problem 2 shown in Table 1. In this case, we find three different equilibria: (π_{A1}, π_{B2}) with payoffs (15,14) and delays (3,2) for agent A and B, respectively; another NE is (π_{A2}, π_{B1}) , with payoffs (14,15) and a delay of (2,3) time steps, respectively; the last NE is a mixed strategy with probabilities 0.001 and 0.999 for π_{A1} and π_{A2} of agent A, and probabilities 0.001 and 0.999 for strategies π_{B1} and π_{B2} of agent B. In this problem we have a cell with $-\infty$ as payoff of the two agents. This payoff represents that there does not exist a valid joint schedule for the plans due to an unsolvable conflict as the one shown in Figure 3.

Table 3: Problem 2

In conclusion, our approach simulates how agents behave with several strategies and it returns an equilibrium solution that is stable for all of the agents. All agents participate in the schedule profile solution and their utilities are dependent on the strategies of the other agents regarding the conflicts that appear in the problem.



	π_{B1}	π_{B2}	π_{B3}	π_{B4}
π_{A1}	$-\infty, -\infty$	7.5,9 (3,2)	18.2 (0,2)	14.5,9 (1,0)
π_{A2}	9,7.5 (2,3)	9,9 (2,2)	16,-1.5 (0,3)	16,9 (0,0)
π_{A3}	8,11 (0,2)	8,16 (0,0)	8,2 (0,2)	8.5,5 (0,1)
π_{A4}	2,18 (2,0)	-1.5,16 (3,0)	9,9 (0,0)	5.5,9 (1,0)

Conclusions and future work

As for future work, we intend to explore two different lines of investigation. The exponential cost of this approach represents a major limitation for being used as a general MAP method for self-interested agents. Our combination of a general+internal game can be successively applied in subproblems of the agents. Considering that this approach solves a subset of goals of an agent, the agent could get engaged in a new game to solve the rest of his goals, and likewise for the rest of agents. Then, a MAP problem can be viewed as solving a subset of goals in each repetition of the whole game. In this line, the utility functions of the agents can be modeled not only to consider the benefit of the current schedule profile but also to predict the impact of this strategy profile in the resolution of the future goals. That is, we can define payoffs as a combination of the utility gained in the current game plus an estimate of how the joint plan schedule would impact in the resolution of the remaining goals.

Acknowledgments

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